DenseNet and the Idea of Reusing Features

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DenseNet and the Idea of Reusing Features

SWIPE ->

ResNet connects layers using shortcut connections.

DenseNet takes feature reuse a step further.

In a DenseNet block, each layer receives the feature maps produced by earlier layers and adds its own new feature maps to the collection.

In simplified form:

$x_l = H_l([x_0, x_1, ..., x_{l-1}])$

Here, the brackets represent concatenation of earlier feature maps.

Why is this interesting?

A later layer doesn't have to rely only on the output of the immediately preceding layer.

It can directly access features learned earlier in the network.

This promotes feature reuse and provides short paths for information and gradients.

DenseNet can also be parameter-efficient relative to some conventional architectures because each layer can produce a relatively small number of new feature maps.

But there is a trade-off.

Concatenating feature maps can increase memory usage and make implementation or deployment more demanding.

This is why I wouldn't choose an architecture based on accuracy alone.

I'd also compare parameter count, memory requirements, inference latency, and performance on the actual dataset.

In deep learning, a clever architecture still has to fit the engineering constraints.

#DenseNet #DeepLearning #AIEngineering

Thanks for reading.

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Code: github.com/yasirusman85/Tank